Rosa Silva Hernandez

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Education

New York University, MS in Biomedical Engineering — Expected May 2027

- GPA: 3.7/4.0
- **Coursework:** Physiological Signal Processing, Systems & Computational Simulation, Rehabilitation Engineering

San Jose State University, BS in Biomedical Engineering — Aug 2019-May 2024

- Dean's Scholar (F'23, S'24)

Publications

NU Rajesh..., **RA Hernandez**, ... et al. *3D-Printed Latticed Microneedle Array Patches for Tunable and Versatile Intradermal Delivery*. *Advanced Materials*, 2024.

Experience

Research Assistant, Jurney Lab, San Jose State University – San Jose, CA — Aug 2023– May 2024

- Developed MATLAB pipelines for quantifying mitochondrial morphology from microscopy images, automating analysis and reducing manual quantification by 90%.
- Built digital segmentation protocols (Photoshop, ImageJ, MATLAB) to extract morphological features, replacing live-cell staining with a reproducible computational workflow.

Stanford Undergraduate Research Fellow (SURF), DeSimone Lab – Palo Alto, CA — June 2023 – Aug 2023

- Designed 3D-printed latticed microneedle array patches (L-MAPs) in Fusion 360, optimizing insertion mechanics and drug delivery efficiency for intradermal therapeutics.
- Fabricated and characterized microneedle patches using Carbon CLIP 3D printing, Instron mechanical testing, and Optical Coherence Tomography (OCT) to validate penetration depth and structural integrity.
- Co-authored a publication in *Advanced Materials* and received a Predoctoral Poster Award at Stanford's Cardiovascular Research Symposium (CVRS).

Lead Undergraduate Researcher, Bellofiore Cardio Lab – San Jose, CA — May 2022 – May 2023

- Engineered a transparent silicone test section with a matched refractive index to enable laser-based visualization of thrombogenic flow through mechanical heart valves.
- Designed and fabricated three prototype molds using 3D printing and developed a silicone curing protocol, enabling reproducible manufacturing of experimental flow chambers.

Projects

DHEA-S Deconvolution & State Estimation, NYU — 2025

- Developed a MATLAB pipeline to estimate adrenal DHEA-S secretion dynamics from 24-hour subcutaneous measurements using physiological state-space modeling, deconvolution, and marked point process state estimation.
- Performed batch analyses on the NYU HPC cluster across 11 age-matched participant pairs, achieving model fits of $R^2 > 0.8$ for 9 of 11 datasets.

Honors & Awards

- NYU Academic Scholarship (2025)
- Predoctoral Poster Award — Cardiovascular Research Symposium (2023) — Stanford University
- Stanford Undergraduate Research Fellowship (2023)

Skills

Programming: MATLAB, Python, R, HPC

Engineering: Fusion 360, CAD, Carbon CLIP 3D Printing, Instron Mechanical Testing, OCT, Microscopy, ImageJ

Research & Analysis: Signal Processing, State Estimation, Statistical Analysis, Scientific Writing, Experimental Design, Device Prototyping